



Innovative and visionary Senior Data Engineer with 9 years of experience architecting high-performance, cost-optimized data platforms across Azure and on-premises ecosystems for top-tier financial institutions. Specializes in transforming complex, multi-source data into automated, highly parameterized ETL pipelines using PySpark, Kafka, and Airflow. Recognized for spearheading massive legacy system migrations—including decommissioning enterprise SAP environments—to drastically reduce licensing costs. Proven track record of integrating GenAI tools (Claude Code) into development workflows to accelerate zero-defect delivery, alongside deep proficiency in designing high-availability databases (Cosmos DB, Databricks) that drive massive query performance improvements.

WORK EXPERIENCE

DATA ENGINEER CONSULTANT

Emirates NBD (ValueLabs)

Jan 2025 - Present

- **Engineered** an end-to-end data pipeline for Wholesale Banking customer deals using PySpark, Kafka, and ODS, standardising approvals and tracking deal expiries to deliver significant business cost savings.
- **Spearheaded** the decommissioning of SAP HANA by migrating existing BDM workflows to PySpark for 5 banking entities, eliminating SAP license costs and successfully automating ~180 daily jobs loading to Oracle.
- **Designed** a highly parameterised PySpark framework configured via simple YAML files, allowing the engineering team to replicate ETL jobs with zero defects rapidly.
- **Managed** complex daily and monthly ETL loads aggregating multiple finance sources and pricing structures, leveraging Airflow, Oozie, and GenAI tools like Claude Code to accelerate deployment timelines without any critical issues.

Senior Data Engineer

EPAM Systems (Client: Canadian Tire)

Dec 2021 - Jan 2025

- **Developed** feature-based data pipelines using Azure Data Factory, enabling a new analytical web application for dealers that seamlessly processed 100 GB of daily data.
- **Architected** the migration of enterprise data from Hive tables into an Azure Cosmos DB application database, reducing query retrieval times by 40%.
- **Collaborated** directly with Business Analysts and Backend Developers to define business requirements and structure optimized JSON schemas for Cosmos DB.

Research and Development Engineer

ABB Global Limited

Feb 2020 - Dec 2021

- **Built** a real-time data pipeline across Azure cloud and on-premises deployments to process semi-structured IoT telemetry data from ActiveMQ and Event Hub.

- **Designed** a framework to deploy Machine Learning models on unstructured data (text, images, and videos), routing predictive results to Cosmos DB and MongoDB for downstream analytics.
- **Developed** automated PySpark batch jobs to clean, transform, and migrate data from the data lake to the curated store, scheduling executions via Rundeck, Azure Pipelines, and Azure Data Factory.
- **Created** custom Python APIs facilitating CRUD operations across various target data systems, improving data accessibility for the engineering team and saving an estimated 15 hours of manual effort per week.

Data Engineer

Infosys (Clients: PwC, Exelon)

Mar 2017 - Feb 2020

- **Developed** PySpark-based ETL workflows to migrate diverse data formats (CSV, JSON, MySQL) into an on-premises HDFS environment in Parquet format.
- **Optimized** data delivery by executing query transformations and creating tables on top of HDFS files to push data to the curated layer, scheduled reliably using Cron.
- **Automated** data API testing by writing robust test scripts using Python and Selenium, successfully validating data loads from Oracle DB to MongoDB.
- **Assisted** in the architectural design of the data warehouse schema and actively resolved bugs via the VSTS DevOps platform.

RELEVANT SKILLS

- **Core Engineering:** Python, PySpark, SQL/NoSQL, Data APIs, Shell Scripting, Kubernetes.
- **Cloud & Big Data:** Azure Data Factory, Azure Databricks with Delta Lake and Unity Catalog, Azure Cosmos DB, Hadoop, Spark (SQL & Kafka/ Flink Streaming).
- **Modern Data Stack & Orchestration:** Snowflake, dbt, Apache Airflow, Oozie.
- **Advanced Pipelines:** Real-time Streaming (Kafka/Flink), LLM/GenAI Pipelines (Claude Code), Data Governance

PERSONAL PROJECTS

- **Vasaicoronaresources.net:** Designed and engineered a highly available web application using the Flask framework to curate essential COVID-19 resources for the Vasai-Virar city region. Built a robust backend to successfully handle traffic spikes of up to 5,000 daily users without downtime.

EDUCATIONAL HISTORY

**Bachelor of Engineering in
Electronics & Telecom**
Mumbai University,
2012 - 2016

CERTIFICATIONS

- Azure Developer Associate
- Azure IoT Developer Specialty